

Faculty of Science, Engineering and Built Environment

SEE216 Analogue and Digital Electronics

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

This is a knowledge and skill-based unit focusing on analogue and digital electronics. The first part of the unit covers the theory and applications of combinatorial and sequential logic circuits and systems. These will include standard SSI and MSI devices such as logic gates, flip-flops, counters and adders. The second part of the unit covers the operation and applications of amplifiers and related devices. It will cover small-signal amplifiers based on bipolar-junction transistors and junction field effect transistors, as well as practical op-amps and its applications.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Benjamin Champion
Campus Geelong Waurn Ponds
Email: benjamin.champion@deakin.edu.au
Phone: +61 3 522 73772
Other members of the team and how to contact them

Other members of the unit team will be announced on the unit site at the start of the trimester.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- Seminar questions were very helpful.
- Seminar and studio sheets are well–structured and comprehensive .
- Seminars on the digital aspects were succinct and easy to follow.
- Resources were available from the start.
- Accessing the wide range of detailed and informative resources provided.
- How the unit flowed by each topic being an extension of the previous.
- How thoroughly each topic was covered.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Continuing to emphasise the seminars and the content available in them.
- Assessments have been enhanced to ensure that it is easy to understand the work that students needed to do.
- All content will be available very early in the teaching trimester.
- The practical tasks have been enhanced to ensure that each student can get the most out of the limited practical sessions.
- Some improvements in the seminar sessions and notes to improve flow for ease of understanding.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Design and build combinational and sequential logic circuits that map given inputs to achieve desired output patterns.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving
ULO2	Use number conversion and perform binary arithmetic that mimics the operation of a digital computer.	GLO1: Discipline-specific knowledge and capabilities
ULO3	Apply fundamental technical knowledge to analyse the operation of transistors both as switches and as small-signal amplifiers.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving
ULO4	Apply fundamental technical knowledge to describe and analyse op-amp circuits.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 50% in the End-of-Unit Assessment and a mark of at least 50% on the practical experience (laboratory) reports.

Brief summary of the hurdle requirement	Rationale
To be eligible to obtain a pass in this unit, students must achieve an overall mark of at least 50% in the End-of-Unit assessment and a mark of at least 50% on the practical experience (laboratory) reports.	Students are required to demonstrate their basic knowledge of analogue and digital electronics through both practical and theoretical applications.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Digital circuit analysis (Problem solving task)**
Brief description of assessment task	The assessment task will involve solving problems associated with the analysis and design of logic circuits.
Detail of student output	An individual written assessment task which involves problem solving. The assignment has approximately 4 problems (questions) with various degree of difficulties. Students are required to provide effective solutions to these problems. The problems will be solved on paper and submitted electronically on the unit site. A Dropbox will be created for the assessment.
Grading and weighting (% total mark for unit)	20% Marked and Graded
This task assesses your achievement of these Unit Learning Outcome(s)	The assignment assesses your technical knowledge related to digital circuits. In particular, combinational and sequential logic circuits and programmable logic devices. The assignment's questions are designed to specifically test your understanding of the above technical knowledge. After completion of this assessment task, students will be able to demonstrate: ULO1: through their design of combinational and sequential logic circuits. ULO2: though their use of number conversion and binary arithmetic method.
This task assesses your achievement of these Graduate Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate: GLO1 (Discipline knowledge): through the explanation of the concepts and knowledge to analyse and design digital circuits. GLO4 (Critical thinking): through the explanation of the reasons behind their decisions made for designing logic circuits. GLO5 (Problem Solving): through their design of logic circuits.
How and when you will receive feedback on your work	Students will receive feedback via the unit site in the form of a mark with comments and feedback, normally within 2 weeks of the submission close date. In addition, full working solutions to the problems will also be provided a few days after the closing of the assignment submission.
When and how to submit your work	Submit electronically via the unit site. **Due date: End of Week 5 - Friday 4 April 2025, 8pm (AEST)**

Summative assessment task 2

Title	**Analogue circuit analysis (Problem solving task)**
Brief description of assessment task	The assessment task will involve solving problems associated with the analysis and design of analogue circuits.
Detail of student output	An individual written assessment task which involves problem solving. The assignment has approximately 4 problems (questions) with various degree of difficulties. Students are required to provide effective solutions to these problems. The problems will be solved on paper and submitted electronically on the unit site. A Dropbox will be created for the assessment.
Grading and weighting (% total mark for unit)	20% Marked and Graded

This task assesses your achievement of these Unit Learning Outcome(s)	The assignment assesses your technical knowledge related to analogue circuits. In particular, transistor circuits as small-signal amplifiers and switches, and operational amplifier circuits. The assignment's questions are designed to specifically test your understanding of the above technical knowledge. After completion of this assessment task, students will be able to demonstrate: ULO3: through their analysis of transistor circuits to process analogue signals ULO4: through their analysis of op-amp circuits to process analogue signals
This task assesses your achievement of these Graduate Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate: GLO1 (Discipline knowledge): through the explanation of the concepts and knowledge to design analogue circuits. GLO4 (Critical thinking): through the explanation of the reasons behind their decisions made for designing analogue circuits. GLO5 (Problem Solving): through their design of analogue circuits.
How and when you will receive feedbackon your work	Students will receive feedback via the unit site in the form of a mark with comments and feedback, normally within 2 weeks of the submission close date. In addition, full working solutions to the problems will also be provided a few days after the closing of the assignment submission.
When and how to submit your work	Submit electronically via the unit site. **Due date: End of Week 11 - Friday 23 May 2025, 8pm (AEST).**

Summative assessment task 3

Title	**Laboratory report**
Brief description of assessment task	Practical reports on the laboratory exercises based on digital and analogue circuits. For campus students, practicals will take place in the labs at scheduled sessions on multiple days during the trimester. Students will work in a small group and they will need to work, learn and share data effectively in a team in order to effectively carry out the practical assessments in a timely manner. Online students will carry out the laboratory tasks in a single day intensive session.
Detail of student output	Students will provide written reports on the lab exercises, explaining the results obtained. This is an individual assessment task.
Grading and weighting (% total mark for unit)	20% Graded
This task assesses your achievement of these Unit Learning Outcome(s)	The practical report assesses your practical skills that are very important for an engineer. It also assesses your technical knowledge such as Boolean algebra, basic gates, flip flops, and transistor amplifier circuits. After completion of this assessment task, students will be able to demonstrate: ULO1: through their observation of the workings of flip-flops in a ripple counter. ULO2: through their observation of the workings of a 4-bit adder. ULO3: through their analysis of an amplifier circuit and comparison of theoretical and measured values.
This task assesses your achievement of these Graduate Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate: GLO1 (Discipline knowledge): through the explanation of the concepts and knowledge to design digital and analogue circuits. GLO4 (Critical thinking): through the explanation of the reasons behind their decisions made for designing digital and analogue circuits. GLO5 (Problem solving): through their design of digital and analogue circuits.
How and when you will receive feedback on your work	Students will receive feedback via the unit site in the form of a grade with comments and feedback, normally within 2 weeks of the submission close date.
When and how to submit your work	Submit electronically via the unit site. **Due date: End of Week 10 - Friday 16 May 2025, 8pm (AEST)**

Summative assessment task End-of-unit assessment or Examination

Title	**End of unit assessment - Written**
Brief description of assessment task	This is an individual assessment task. The assessment task will involve solving problems associated with the analysis and design of both digital and analogue circuits, covering all topics in the unit.
How you demonstrate your achievement	Students are required to undertake a 2-hour timed release assessment. A written assessment is a form of assessment that asks you to use text to communicate your knowledge, skill or understanding of a topic. It is completed online. At a set release time, you will be given the assessment instructions and will complete the assessment by a set deadline.
Grading and weighting (% total mark for unit)	40%
This task assesses your achievement of these Unit Learning Outcome(s)	This assessment task assesses your achievements of ULO1-4. Students will demonstrate: ULO1: through their design of combinational and sequential logic circuits. ULO2: though their use of number conversion and binary arithmetic method. ULO3: through their analysis of transistor circuits to process analogue signals ULO4: through their analysis of op-amp circuits to process analogue signals
This task assesses your achievement of these Graduate Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate: GLO1 (Discipline knowledge): through the analysis and explanation of the concepts and knowledge of the operations of digital and analogue circuits.
How and when you will receive feedback on your work	Deakin University will release the final assessment results at the stipulated timeframe. Students will receive a mark, which is an indicator of their overall performance in this unit of study. Students may request feedback on their performance in this task.
When and how to submit your work	Students will be required to undertake a timed online assessment during the End of unit assessment Period. You will have a start time (e.g., 9am) with a specified 2-hour time limit. Further details will be provided on the unit site. It is the responsibility of students to review their End of unit assessment timetable when it is released via StudentConnect. Further information including timetable release dates is available at end-of-unit assessments and examinations.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour seminar per week, 3 x 2 hour practical experience (laboratory) per trimester.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 1 hour seminar per week. Students are required to attend and participate in practical experiences (laboratories) at the Waurn Ponds (Geelong) Campus for the scheduled sessions during the trimester intensive activities.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

Learning and assessment activities in this unit for campus students includes watching recorded content, designated on campus activities, assessment tasks, readings and study time. For online students the time should be divided between online learning activities, watching recorded content, discussion boards, designated activities, assessment tasks, readings and study

time.

Students are also expected to spend time reading the practical resources provided and completing self-directed problem-based exercises and assessment tasks.

For online students, the seminar will be run online and the practical experiences (laboratories) will be held at the Geelong Waurn Ponds Campus on the scheduled 1 day during the trimester intensive week.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE216 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

There are no prescribed textbooks for this unit.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.
Recommended learning resources for this unit are:

- R.J. Tocci, *Digital Systems: Principles and Applications*, 12th global edition, Prentice-Hall, Englewood Cliffs, N.J., 2018.
- Floyd, T.L., *Electronic Devices*, 10th edn, Prentice Hall, 2018.

Students are not required to purchase these texts, they are available in the University Library.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment due date
1	3 March 2025	Introduction to digital systems and number systems	
2#	10 March 2025	Logic circuits and Boolean algebra	
3	17 March 2025	Latches and flip-flops	
4	24 March 2025	Binary arithmetic and binary adders	
5	31 March 2025	Electrical characteristics and integrated circuits	**Assessment 1: Digital circuit analysis (problem solving task)** due 4 April
6	7 April 2025	Bipolar Junction Transistor bias circuits	
7*~	14 April 2025	Small signal BJT amplifiers	
8	28 April 2025	BJT amplifier frequency response	
9	5 May 2025	Junction Field-Effect Transistor bias circuits	
10	12 May 2025	Small-signal JFET amplifiers	**Assessment 3: Laboratory reports** due 16 May
11	19 May 2025	Operational Amplifiers	**Assessment 2: Analogue circuit analysis (problem solving task)** due 23 May
12	26 May 2025		Study period
13-14^	2 June 2025		**End-of-unit assessment**

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed